Example 6 Find the side of the equilateral triangle of perimeter 129 *cm*.

Solution Perimeter of the equilateral triangle, P = 129 *cm*

$$a + a + a = 129$$
$$3 \times a = 129$$
$$a = \frac{129}{3}$$
$$= 43\ cm$$

The side of the equilateral triangle is 43 *cm*.

Example 7 Thendral, Tharani and Thanam are given a thread piece each of length 12 *cm*. They are asked to make a rectangle, a square and a triangle respectively with the thread for their Math activity. In how many ways, can they make the respective shapes?

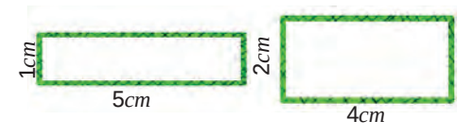


Solution

Thendral

Perimeter of the rectangle, P = 12 *cm*

$$2(l + b) = 12$$
$$l + b = \frac{12}{2} = 6\ cm$$



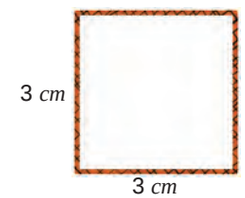
The possible pairs of measures whose sum is 6 are (5,1) and (4, 2).

Hence, Thendral can make a rectangle in 2 ways. She can make a rectangle of length 5 *cm* and breadth 1 *cm* and another one with length 4 *cm* and breadth 2 *cm*.

Tharani

Perimeter of the square, P = 12 *cm*

$$4 \times s = 12$$
$$s = \frac{12}{4} = 3\ cm$$



Hence, Tharani can make only one square of side 3 *cm*.

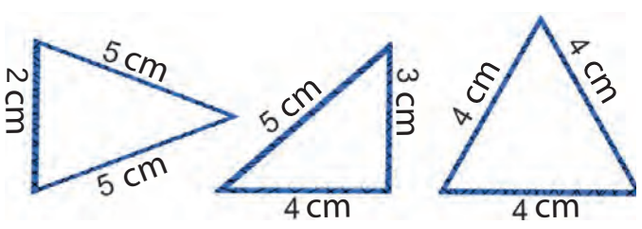
Thanam

Perimeter of the triangle, P = 12 *cm*

$$a + b + c = 12\ cm$$

The possible triplets of measures whose sum is 12 and also satisfying the triangle inequality are (2, 5, 5) ; (3, 4, 5) ; (4, 4, 4).

Hence, Thanam can make 3 triangles of sides 2 *cm*, 5 *cm* & 5 *cm*; 3 *cm*, 4 *cm* & 5 *cm* and 4 *cm*, 4 *cm* & 4 *cm*.



Think

Can different shapes have the same perimeter?

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